

Numerical Solution of the Two-Dimensional Poisson Equation

Using a Five-Point Finite Difference Method

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1 Introduction

The purpose of this project was to move from a one-dimensional finite-difference problem to a two-dimensional elliptic PDE and to see how the discretization leads to a sparse linear system. I used the Poisson equation on a square domain because the model is simple enough to verify with an exact solution, while still introducing several ideas that are important in scientific computing: two-dimensional grids, sparse matrices, linear solvers, convergence, and computational cost.

The code was written in MATLAB. I first checked the numerical solution on moderate grids, then carried out a grid-refinement study up to $N = 161$ points in each coordinate direction. I also measured the solution time over repeated runs and inspected the sparsity pattern of the finite-difference matrix.

2 Mathematical Model

I considered the Poisson equation

$$-\Delta u = f \quad \text{in } \Omega = (0, 1) \times (0, 1), \quad (1)$$

with homogeneous Dirichlet boundary conditions

$$u = 0 \quad \text{on } \partial\Omega. \quad (2)$$

To make the numerical error easy to measure, I chose the exact solution

$$u(x, y) = \sin(\pi x) \sin(\pi y). \quad (3)$$

Since

$$u_{xx} = -\pi^2 \sin(\pi x) \sin(\pi y), \quad u_{yy} = -\pi^2 \sin(\pi x) \sin(\pi y),$$

the corresponding right-hand side is

$$f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y). \quad (4)$$

The exact solution is zero on all four sides of the unit square, so it is consistent with the imposed boundary condition.

3 Finite Difference Discretization

I used a uniform Cartesian grid with spacing

$$h = \frac{1}{N-1}.$$

At an interior point (x_i, y_j) , the standard centered approximations for the second derivatives give the five-point stencil

$$-\Delta u(x_i, y_j) \approx \frac{4u_{i,j} - u_{i+1,j} - u_{i-1,j} - u_{i,j+1} - u_{i,j-1}}{h^2}. \quad (5)$$

Therefore, each interior node produces one linear equation. After arranging the interior grid values into a vector, the full discrete problem can be written as

$$A\mathbf{u} = \mathbf{b}. \quad (6)$$

The matrix A is sparse because each interior unknown interacts only with itself and its four nearest neighbors. In the MATLAB implementation, I assembled the two-dimensional operator from one-dimensional second-difference matrices using sparse Kronecker products, and then solved the resulting system with MATLAB's backslash operator.

4 Numerical Solution

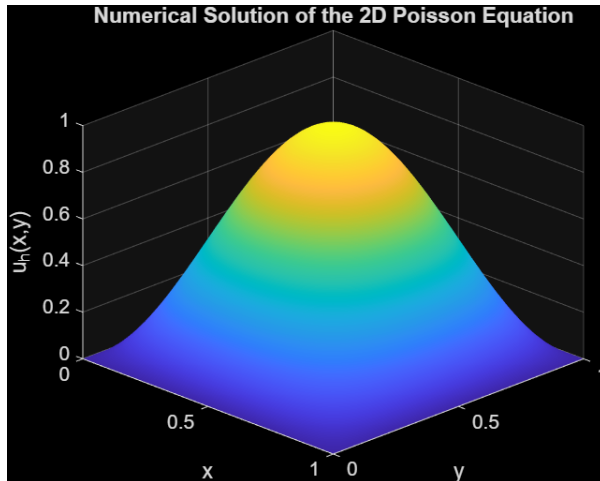
I first used $N = 41$, which gives $39 \times 39 = 1521$ interior unknowns and a grid spacing of $h = 0.025$. The computed infinity-norm error was

$$\|u_h - u\|_\infty = 5.142005 \times 10^{-4},$$

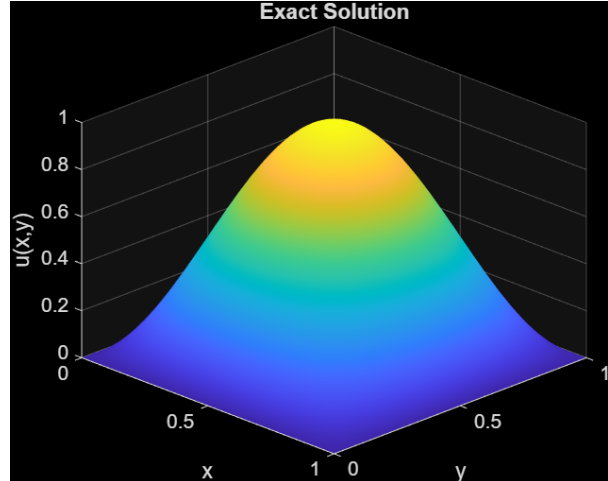
and the discrete L^2 error was

$$\|u_h - u\|_{2,h} = 2.571002 \times 10^{-4}.$$

Figure 1 compares the numerical and exact surfaces. Visually, the two surfaces are almost indistinguishable at this resolution.



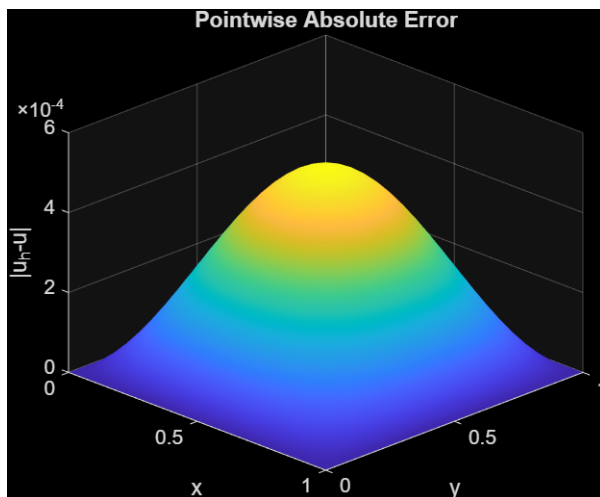
(a) Numerical solution, $N = 41$.



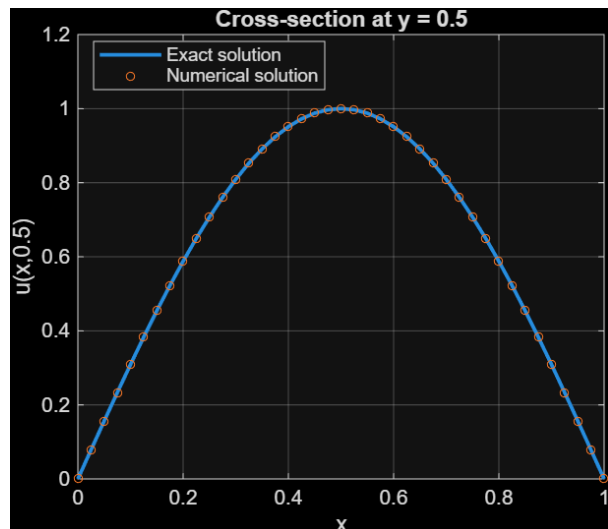
(b) Exact solution.

Figure 1: Numerical and exact solutions for the two-dimensional Poisson problem.

The pointwise absolute error is shown in Figure 2. The error is zero on the boundary because the boundary values are imposed exactly, while the largest error occurs near the center of the domain where the solution itself has its maximum.



(a) Pointwise absolute error for $N = 41$.



(b) Cross-section at $y = 0.5$.

Figure 2: Error distribution and a one-dimensional cross-section through the computed solution.

I then refined the grid to $N = 81$. The number of interior unknowns increased to $79^2 = 6241$, while the infinity-norm error decreased to

$$1.285204 \times 10^{-4}.$$

This is approximately one fourth of the $N = 41$ error, which is an early indication of second-order convergence.

5 Grid Convergence Study

To check the convergence rate more systematically, I used five grids: $N = 11, 21, 41, 81$, and 161 . The results are listed in Table 1.

Table 1: Grid-refinement results for the five-point finite-difference method.

N	h	Interior unknowns	L^∞ error	Observed order
11	0.10000	81	8.2654×10^{-3}	–
21	0.05000	361	2.0587×10^{-3}	2.0053
41	0.02500	1521	5.1420×10^{-4}	2.0013
81	0.01250	6241	1.2852×10^{-4}	2.0003
161	0.00625	25281	3.2128×10^{-5}	2.0001

The observed order was computed from successive errors using

$$p = \frac{\log(E_h/E_{h/2})}{\log 2}. \quad (7)$$

The values approach 2 very closely as the grid is refined. This agrees with the expected second-order accuracy of the centered finite-difference approximation. The log-log convergence plot in Figure 3 also shows an almost straight line.

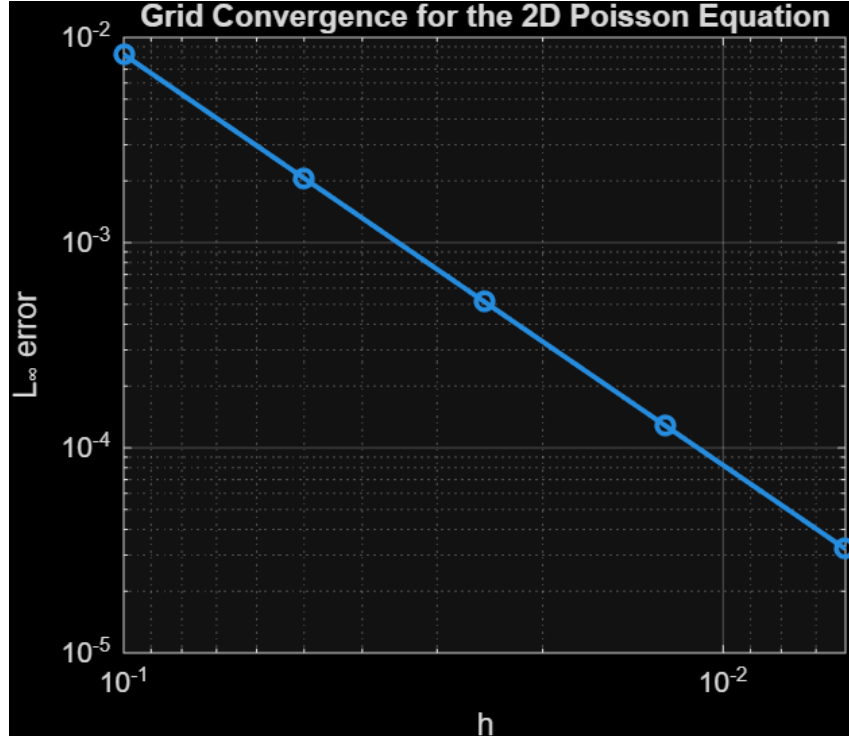


Figure 3: Infinity-norm error versus grid spacing.

6 Computational Cost and Sparse Structure

For a two-dimensional problem, refining the grid increases the number of unknowns quickly. When h is roughly halved, the number of interior unknowns grows by about a factor of four. To get a simple picture of the computational cost, I ran each grid size ten times and recorded the average elapsed time. The results are shown in Table 2.

Table 2: Average runtime over ten runs for each problem size.

N	Interior unknowns	Average time (s)
11	81	5.1040×10^{-4}
21	361	3.5010×10^{-4}
41	1521	1.2317×10^{-3}
81	6241	5.9271×10^{-3}
161	25281	2.91851×10^{-2}

The two smallest problems are so inexpensive that fixed MATLAB overhead and timing noise are comparable to the actual solve time, so the first two measurements are not monotone. From $N = 41$ onward, however, the growth in runtime is clear as the system size increases. Figure 4 shows the same trend on logarithmic axes.

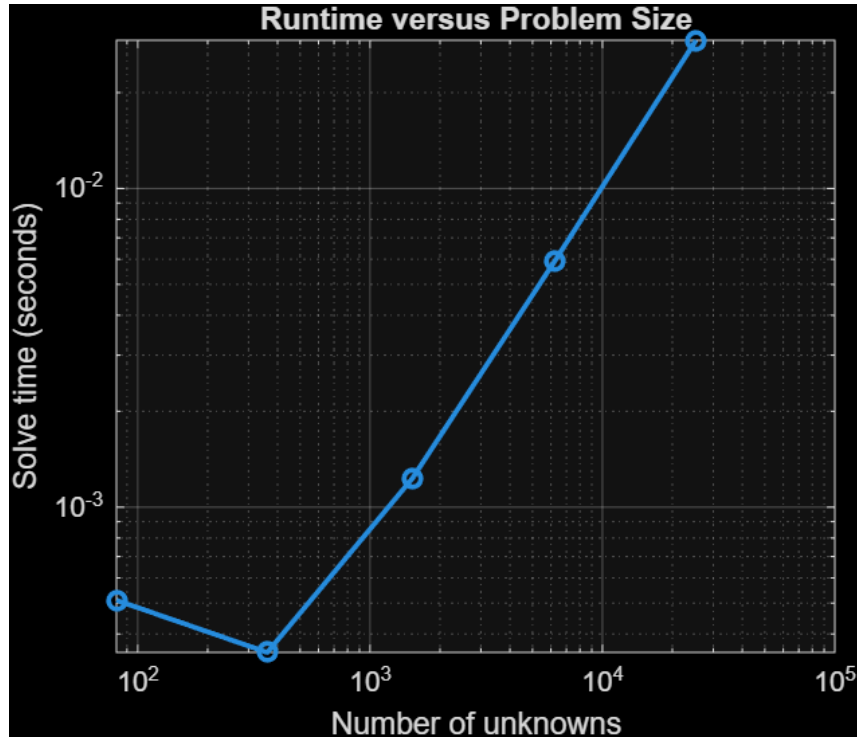


Figure 4: Average runtime versus the number of interior unknowns.

The sparse structure is also important. For $N = 41$, the system has 1521 unknowns. A dense 1521×1521 matrix would contain

$$1521^2 = 2,313,441$$

entries. The actual finite-difference matrix has only 7449 nonzero entries, or about 0.32% of the entries in the corresponding dense matrix. Figure 5 shows the sparsity pattern.

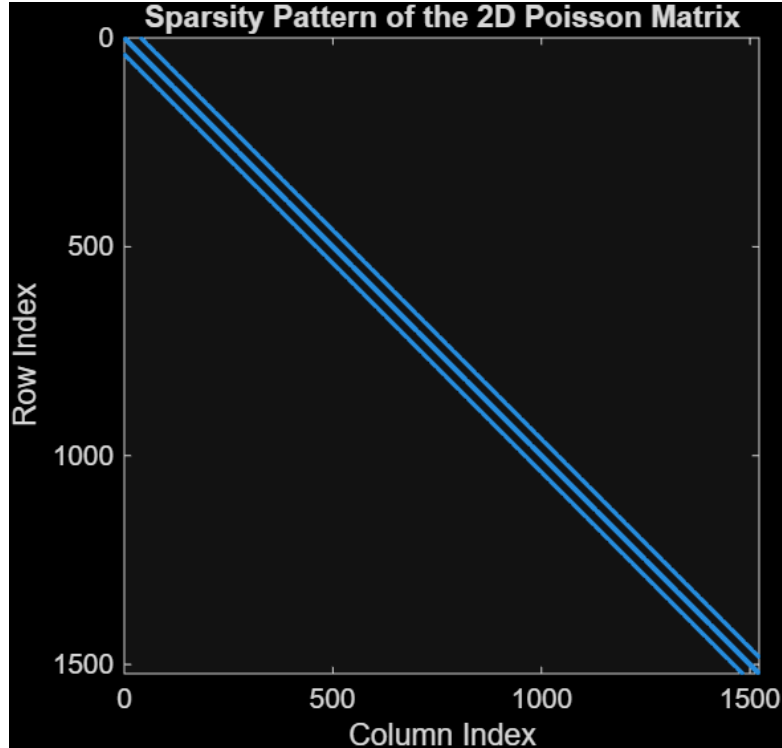


Figure 5: Sparsity pattern of the two-dimensional Poisson matrix for $N = 41$.

This was one of the most useful parts of the project for me because it makes the connection between the local five-point stencil and the global sparse matrix very clear. It also shows why sparse storage and sparse linear algebra are essential when PDE grids become large.

7 Conclusion

In this project, I solved a two-dimensional Poisson equation on the unit square using the standard five-point finite-difference method. The numerical solution agreed closely with the known exact solution, and the grid-refinement study produced an observed convergence rate essentially equal to two. The experiments also showed how quickly the number of unknowns and the computation time grow as the grid is refined.

A second important outcome was seeing how the finite-difference stencil leads naturally to a sparse matrix. Even for the moderate $N = 41$ grid, fewer than one percent of the entries in the corresponding dense matrix are nonzero. This is the part of the project that connects most directly to larger scientific-computing problems.

A natural next step would be to compare direct and iterative sparse solvers, and then study how the computation can be parallelized for larger grids.